Hyperbolic PDEs

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Hyperbolic PDE (IBVP as time is involved). Both hyperbolic & parabolic are IBVP, elliptic is BVP.

Consider the wave equation $\dfrac{\partial^2 u}{\partial t^2} = \dfrac{\partial^2 u}{\partial x^2}$ $\rightarrow (1)$

Subject to I.C. $u(x,0) = f(x)$

$\dfrac{\partial u}{\partial t}(x,0) = g(x)$, $-\infty < x < \infty$

B.C. $u(0,t) = u(1,t) = 0$; $t > 0$

Consider difference approx. $\rightarrow$ $\dfrac{\partial^2 u}{\partial t^2} = \dfrac{1}{k^2}\delta_t^2 u + O(k^2)$

and $\dfrac{\partial^2 u}{\partial x^2} = \dfrac{1}{h^2}\delta_x^2 u + O(h^2)$

Applying approx. to (1) at the nodal point (mh, nk), we get

$$\frac{1}{k^2}\delta_t^2 U_m^n = \frac{1}{h^2}\delta_x^2 U_m^n$$

$$\frac{1}{k^2}\left(U_m^{n+1} - 2U_m^n + U_m^{n-1}\right) = \frac{1}{h^2}\left(U_{m+1}^n - 2U_m^n + U_{m-1}^n\right)$$

$$U_m^{n+1} = 2(1-r^2)U_m^n + r^2\left(U_{m+1}^n + U_{m-1}^n\right) - U_m^{n-1} \rightarrow (2)$$

where $r = \dfrac{k}{h}$

This is an explicit 3 level formula.

Note that the min. no. of time levels that can be used in a hyperbolic problem is 3.

In order to start computations, we need the data on 2 lines ($t = 0$ and $t = k$).

The information required at $t = k$ is obtained using a suitable approximation to the initial condition $\dfrac{\partial u}{\partial t}(x,0) = g(x)$

If we use the approx. (2nd order, central diff.)

$$\frac{\partial u}{\partial t} = \frac{1}{2k}\left(u_m^{n+1} - u_m^{n-1}\right) + O(k^2)$$

we get $u_m^{-1} = u_m^{1} + 2k\, g_m$ (putting $n=0$) — ③

The extraneous point u_m^{-1} introduced in ② for $n=0$ are removed by using ③.

LTE of ② $\rightarrow$ $O(k^2 + h^2)$

Von Neumann stability gives the condition $r \leq 1$ for the stability of the scheme.

Note This stability condition for the explicit formula is known as CFL (Courant Friedrich Levy) condition.

If we replace u_m^n by a weighted sum

$$\theta\, U_m^{n+1} + (1-2\theta)\, U_m^n + \theta\, U_m^{n-1} \text{ in the RHS of}$$

$$\frac{1}{k^2}\delta_t^2 U_m^n = \frac{1}{h^2}\delta_x^2 U_m^n$$

then we get

$$\delta_t^2 U_m^n = r^2 \delta_x^2\left[\theta\, u_m^{n+1} + (1-2\theta)\, U_m^n + \theta\, u_m^{n-1}\right] \quad -④$$

$$(0 \leq \theta \leq 1)$$

The diff. scheme is of order $O(h^2 + k^2)$

Stability condition 1) $\theta \geq \frac{1}{4}$ given unconditional stability

2) $0 < \theta < \frac{1}{4}$, $0 < r^2 \leq \frac{1}{1-4\theta}$ (conditional stability)

Consider two dimensional wave equation

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x_1^2} + \frac{\partial^2 u}{\partial x_2^2} \rightarrow (5)$$

I.C. : $u(x_1, x_2, 0) = f(x_1, x_2) \quad 0 \le x_1, x_2 \le 1$

$\frac{\partial u}{\partial t}(x_1, x_2, 0) = g(x_1, x_2) \quad 0 \le x_1, x_2 \le 1$

B.C. : -

An explicit scheme can be written as

$$\delta_t^2 u^n_{l,m} = r^2 \left(\delta_{x_1}^2 + \delta_{x_2}^2 \right) u^n_{l,m}$$

or
$$u^{n+1}_{l,m} = 2(1 - r^2) u^n_{l,m} + r^2 \left(u^n_{l-1,m} + u^n_{l+1,m} + u^n_{l,m-1} + u^n_{l,m+1} \right) - u^{n-1}_{l,m}$$

Order of the method $= O(k^2 + h^2)$

Stability condition $= 0 < r \le \frac{1}{\sqrt{2}}$

Implicit scheme for (5) can be obtained by generalising implicit scheme (4) of 1 dimension.

$$\delta_t^2 u^n_{l,m} = r^2 \left[\theta \left(\delta_{x_1}^2 + \delta_{x_2}^2 \right) u^{n+1}_{l,m} + (1 - 2\theta) \left(\delta_{x_1}^2 + \delta_{x_2}^2 \right) u^n_{l,m} + \theta \left(\delta_{x_1}^2 + \delta_{x_2}^2 \right) u^{n-1}_{l,m} \right]$$

Stability analysis 1) $\theta \ge \frac{1}{4}$ $\rightarrow$ Unconditionally stable

2) $0 < \theta < \frac{1}{4}$ $\Rightarrow$ $r^2 \le \frac{1}{2(1 - 4\theta)}$ (conditionally stable)